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Virtual vs. human influencer: Effects on users' perceptions and brand outcomes

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ABSTRACT

In the recent wave of artificial intelligence-based innovations, Virtual influencers (VIs) are one of the newest and most promising innovations. Despite their indisputable potential, still little is known about users' reactions to non-human endorsers. Our experimental study investigates users' perceptions of VIs engaged in sponsorship activities on Instagram, examining the impact of the influencer's nature (human vs. virtual) and possible interaction effects of the type of message conveyed (hedonic vs. utilitarian) on the influencer and brand-related variables. In line with the idea that individuals apply human social rules when interacting with computers, as established in the CASA paradigm, our results from an online sample (N = 238) show that disclosing the synthetic nature of a fully anthropomorphized virtual persona does not affect influencer and brand perceptions. Despite virtual influencers' objective lack of capacity to perceive and feel affective and sensorial information, their acceptance by the users is not hindered by an emotion-based communicative style. The research provides social and business implications for companies aiming to implement virtual influencers successfully.

1. Introduction

In 2018 Time magazine issued its '25 Most Influential People on the Internet', a roundup of individuals or groups that were able to generate an outstanding global impact on social media and digital communication throughout the year [1]. Among the featured personalities, many lifestyle and fashion influencers made their way to the top, including the Instagram model Miquela Souza. With a net worth that has been estimated to range from 10 to 16 million dollars [2], Lil Miquela shares many features with her lifestyle guru peers: from the millions of followers to the dreamlike, spotless aesthetics of her Instagram feed. Except for the fact that, unlike the other personalities ranked, Miquela is not *real.* Although today's digital platforms are populated by AI-powered and AI-generated content – from algorithm-driven programmatic advertising to creative content, such as posts completely written by AI-writing tools such as ChatGTP – the phenomenon of virtual influencers (VIs) deserves peculiar attention as it has been redefining how virtual and real persons interact and build online bonds [3]. Within all platforms and social media, VIs' popularity is peaking on Instagram probably due to the social media's inherently visual nature [4]. Thanks to advanced Computer-Generated Imagery (CGI) techniques – combined with a strategic use of narrative strategies typical of Instagram storytelling – VIs are more than accurate avatars with different levels of anthropomorphism, human likeness, and personality [5]. As was the case with disembodied agents like chatbots, these artificially generated endorsers and their human counterparts are not only coexisting on the same social media platforms [6] but also engaging in similar ways with their audience. VIs appear as having their own public persona through human-like storylines, they are portrayed in pictures together with humans, they publicly interact with human users in their comment sections [7], and they advocate for political and social causes [8]. Last, but certainly not least, on social media they endorse and showcase a wide range of services and products by partnering up with brands like Gucci, Samsung, and Adidas [9], making them a viable alternative to human influencers (HI) [10].

The aim of introducing AI-generated personalities to a brand's pool

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of online ambassadors is to leverage the undoubted fascination that such innovation brings into a world that is oversaturated with sponsored content [11]. Aside from the tremendous performance that VIs have been achieving in terms of engagement metrics [12] and recent findings depicting VIs as a "sweet escape" from the potentially excessive emotional experience of interacting with other humans [10], some are also pinpointing many potentially advantageous innovative features that they might introduce to the influencer marketing scenario. From an ethical perspective, virtual endorsers were expected to be immune to reputational threats or authenticity issues that are inherently attached to the figure of human endorsers [7,13]. For instance, as VIs only exist in the social media realm and do not have real a private sphere, brands should not be concerned about risking their reputation by entering endorsement deals with individuals that might cause scandal or exhibit questionable behavior elsewhere [14,15]. Second, for evident reasons, they cannot be perceived as publicity stunts motivated by financial gains, a process that might decrease effectiveness, posing a threat to the endorsing brand as well as HIs [16]. Third, having an intrinsic and explicit fake nature, VIs should be unaffected by most authenticity issues that hamper HIs' credibility, such as extremely staged, filtered, or retouched photos [17], thus "*legitimizing VIs' staged fabrication as frankly fake*" [18]. If analyzed through the lens so far adopted for other AI-based services or devices, such as disembodied conversational agents (e.g. chatbots and voice assistants), the interchangeability of virtual and human endorsers - or even the hypothesized utter superiority of the former over the latter in many domains - would however appear as a hype-driven strikingly odd exception to the rule. Indeed, although brand interactions are increasingly involving virtual entities at different touchpoints [19], findings from related research streams suggest a much more nuanced pool of consumer's reactions to AI-based technologies as part of the consumer-brand interaction process [20].

Despite such a wide range of contrasting empirical findings and opposing theoretical debates around anthropomorphized machines, and digital personas more specifically [7,18,21,22], little research has yet explored the potential outcomes of this trend by directly comparing the impact of humans vs. virtual influencers on an all-encompassing pool of both endorser- and brand-related factors [15,18,23]. To the best of our knowledge, while cross-sectional studies assessing social presence toward the virtual influencer and attitude toward the brand exist [24], no experimental study has yet delved into assessing how human vs virtual endorsers impact elements like social presence, trust, and attitude toward the influencer, and trust and attitude toward the brand. As companies currently engage HIs while exploring the transition to virtual influencers, there is keen interest in understanding user perceptions toward the latter [25], and conducting rigorous experimental studies is the optimal and more reliable way to discern differences in user responses between the two and make inferences [26].

Trying to fill this gap, based on marketing scholars' contemporary interest in influencer advertising practices on Instagram [27], we experimentally compared relational perceptions (i.e. social presence and trust) and attitudes generated by Instagram product endorsers. Our study also expands the literature on virtual influencers from a communication-matching perspective. As effective content creation and communication have the potential to considerably boost the success of inbound marketing campaigns [28], virtual influencers, similar to their human counterparts, can play a crucial role in this process [29]. Understanding the type of communication virtual influencers should leverage to be perceived as credible by users is a key aspect that has not yet been studied enough by the scientific community. Thus, to fill this research gap, in addition to examining whether the disclosure of the nature of the influencer (human vs. virtual) affects both the perceptions toward the influencer and sponsored brand, *ceteris paribus*, we also took into consideration possible interaction effects of the nature of the message conveyed by the influencer, comparing emotion-based and function-based captions (hedonic vs. utilitarian).

2. Theoretical framework and hypotheses development

2.1. Users' perceptions toward VIs

While VIs acceptance mechanisms may parallel those in other AI technologies and overlap with those of HIs [30], the widespread adoption of VIs, along with their unique characteristics, necessitates a focused research agenda [29].

In influencer-consumer dynamics, the influencers' capacity to foster a strong parasocial connection with their audience [31] and the users' subjective perception of psychological engagement with others in terms of social presence are pivotal factors [32]. While the personal narratives shared by virtual influencers can help establish emotional connections with their audience [33], recognizing a virtual influencer as a fabricated character could negatively shape people's expectations for both short-term interactions and long-lasting relationships. Initial insights from Stein and colleagues [31] partially addressed this issue and showed that VIs hold a unique position within such a continuum. Despite some aspects of VIs presented personas pose potential challenges in eliciting favorable responses from their audience, they are captivating enough to generate levels of engagement comparable to fully human peers. Byun and Ahn [30] confirmed that VIs can generate levels of social influence akin to HIs in terms of reach, impact, and the cultivation of an intimate bond with followers. Such equivalence may arise from the fact that both human and virtual influencers craft their personas to embody distinct characteristics and adeptly interact with their audiences [31].

Viewing it through the lens of human-computer interaction and the CASA (Computers Are Social Actors) paradigm [34], users' tendency to anthropomorphize machines [35], as well as ascribe social cues to them and expect adherence to social norms [36,37], could be another contributing factor to virtual influencers' success. Within the CASA framework, two major mechanisms determine whether humans evaluate their synthetic counterparts as suitable human-like partners: mindlessness and anthropomorphism [i.a., 36,37]. Both mechanisms are crucial in the specific case of VIs. Mindlessness is inherently linked to Instagram itself (and other social media platforms), rather than to the technology, as this platform is strongly characterized by a habitual, almost ritualized mindless scrolling activity [i.a., 38]. Perceived anthropomorphism has been found to specifically boost consumers' evaluation outcomes in digital agents [70], including virtual influencers, through the activation of social presence and perceived attractiveness [24]. The positive reactions that human-like virtual influencers produce in users, however, suffer a significant halt when the resemblance is high but imperfect [7], a phenomenon known as the Uncanny Valley (UV) effect [39,40]. Anthropomorphized VIs exhibiting cartoon-like features that strongly signal their non-human nature are still stuck at some points within the uncanny valley curve [7]. This makes it difficult to properly disentangle whether the found negative users' reactions are ascribable to the eeriness triggered by this human-like though still imperfect appearance or rather to a much more rooted preference of humans for human actors, *ceteris paribus*. In our study, we try to address this gap by employing a fully humanized character, where the UV is overcome, and the VI's affinity is increased to its highest level of human likeness. The goal is to precisely measure the authentic impact of revealing the virtual nature of the depicted character, focusing on the actual consequences of the disclosure rather than the presence of obvious cues that elicit an uncanny feeling, indicating its synthetic nature. Instagram, as a platform, provides an inherently suitable stage for this purpose due to its prevalent use of manipulated imagery, even in content that narrates human-generated content. The platform offers diverse image editing and modification features, including face filters and augmented reality resources, blurring the lines between what is a purposely fabricated digital persona, and what is a strongly edited real human [41].

To contribute to the knowledge of the evolving relationship between humans and virtual entities and understand how individuals form their attitudes and social connections with such digital entities, we first

compared human vs virtual influencers in terms of social presence, trust, and attitude toward the influencer.

Social presence represents the degree of salience of the "other" person in the interaction, that is feeling the other as psychologically present [42]. Social presence is an important factor for mediated experiences like the ones on social media [43], as well as a precious construct when it comes to improving the lack of human warmth in the online environment [44]. Trust is another "social exchange variable" playing a key role in social interactions, especially in the online context where face-to-face interactions are not possible [45]. Enhancing trust or avoiding a trust erosion reduces the uncertainty and the vulnerability that the users feel, and this is particularly true in the case of AI-enabled features that show an inherent ambiguity [46], as in the instance of VIs. Last but not least, attitude is considered one of the most appropriate dependent variables for those studies involving a fictitious online interaction when actual behavior is not directly measurable. Indeed, a person who holds a favorable attitude towards an action or an object will be more inclined to perform a particular behavior [47].

Thus, drawing on the above-mentioned literature supporting how humans apply the same social heuristics used for human interactions to computers and humanize virtual characters, we hypothesized that.

H1a. There are no significant differences in the users' perception of social presence toward the influencer when comparing a VI vs a HI.

H1b. There are no significant differences in the users' perception of trust toward the influencer when comparing a VI vs a HI.

H1c. There are no significant differences in the users' attitude toward the influencer when comparing a VI vs a HI.

2.2. VIs and brand outcomes

As a second step, we compared human vs virtual influencers in terms of trust and attitude toward the brand, based on the assumption that users' reactions to the VI are likely to extend to the brand that is sponsored by the synthetic endorser. A substantial amount of research highlights that the use of an endorser works as a positive associative link that transfers pleasant feelings, meanings, and attitudes about the endorser to the endorsed brand/product [51,52], potentially leading to increased purchases and consumption [53]. For instance, a study applying automated text analysis to almost 60,000 consumers' comments confirmed that social media influencers facilitate brand engagement through their digital content [54]. By using virtual influencers, brands become part of the influencer's engaging storylines, personalities, and ultimately their emotionally invested communities [7]. Research suggests that VIs are effective in building relevant brand-related variables like brand image and brand awareness [18]. Thomas and Fowler [15] confirm that virtual influencers can generate positive brand benefits similar to those achieved by human celebrity endorsers, while Lou and colleagues [18] provide support for the idea that virtual influencers can be as effective as human celebrity endorsers in shaping favorable brand attitudes.

The endorsement marketing literature indicates the identification with the endorser and the perceived endorser's credibility as two major processes underlying the effect of brand endorsement on advertising effectiveness [48–50]. Extensive literature supports the interconnected nature of trust in both the endorser and the brand, creating a symbiotic relationship where one reinforces the other. This reinforcement is supposed to occur when the users perceive congruence and authenticity between the promoted product, the endorsing influencer, and the brand. Reciprocal spillover effects between influencers and brands undoubtedly exist in terms of relational capital [56].

Although the present study does not delve into analyzing the mutual impacts between virtual influencers and endorsed brands, we recognize that, akin to celebrities and HIs who are seen as relatable and approachable, virtual influencers possess a unique potential to be recognized by consumers as brands advocates. Therefore, it is essential to investigate whether a virtual influencer can be considered as effective as a HI in building relational capital with users. Following the CASA paradigm and recent literature suggesting virtual influencers as likable endorsers facilitating the identification processes [24], as per the attitude toward the brand, we anticipate no differences between virtual and human influencers in generating trustworthiness toward the sponsored brand. Therefore, we hypothesized that.

H2a. There are no significant differences in the users' perception of trust toward the brand when comparing a VI vs a HI.

H2b. There are no significant differences in the users' attitude toward the brand when comparing a VI vs a HI.

2.3. Interaction effects of influencer type and message conveyed

All influencers, including virtual ones, engage with brands by showcasing and promoting a wide range of different products and services. Consistent empirical findings support the idea that the congruence between an endorser and the endorsed product or service is likely to facilitate the meaning transfer and thus participates in building the credibility and trust of the source and contributes to the overall effectiveness of the endorsement [53,57,58]. In the domain of human endorsers, congruence is generally evoked by the alignment between the endorser and the product showcased through the perceived domain of expertise of the influencer [23,48]. Such product-human endorser fit seems to hold equally when leveraging experiential (hedonic) and functional (utilitarian) values of products and services [59] across different purposes, provided that the products or services endorsed are being presented in a way that is perceived congruent with the endorsers' online narratives and overall posting allure [60].

By taking the general product-endorser fit perspective, in the case of VIs, the perceived product-endorser fit may be weakened if the message conveyed focuses on hedonic emotional or sensory values over objective, instrumental ones. In fact, the former are mental states that cannot be directly experienced by AIs, which lack sensory capability [61], and despite nowadays the distinction between humans and AI blurs, AI applications are still commonly perceived as machines rather than human beings [62].

Recently, there has been a growing interest in studying the relationship between the congruence of the product or communication delivered by the virtual influencer and the artificial nature of the virtual influencer [61]. Assuming Gray and Wegner's perspective [63], mind perception consists of two distinct components: agency (perceived capacity to act) and experience (perceived ability to sense and feel). Interestingly, the authors claimed that people usually assign more importance to the ability to experience rather than to agency. As reported in Li and colleagues [61], AI applications possess a high degree of agency (e.g., they are capable of doing tasks autonomously), nevertheless, as they are not actual human beings, consumers cast doubt on their ability to experience due to their incapacity to engage the five senses. The absence of sensory capability of VIs can then foster consumers' skepticism regarding the authenticity of the endorsement information, leading to a decline in perceived trust and attitude in the influencer and the sponsored brand [43].

To sum up, because of VIs' lack of capacity to experience, feel, and perceive, users should prefer humans over virtual agents for transmitting hedonic, emotion-oriented content [64,65]. Rather, users should perceive greater comfort and confidence with virtual agents when they use cognitive-oriented content [65]. Starting from these empirical and theoretical premises, we examined whether leveraging hedonic vs. instrumental values of the showcased product elicits different users' responses in terms of trust and attitude, depending on the influencer's nature, and we hypothesized that.

H3a. The presence of a VI (vs. a HI) leads to higher trust toward the

influencer, in particular when the type of message is utilitarian (vs. hedonic).

H3b. The presence of a VI (vs. a HI) leads to a better attitude toward the influencer, in particular when the type of message is utilitarian (vs. hedonic).

H4a. The presence of a VI (vs. a HI) leads to higher trust toward the brand, in particular when the type of message is utilitarian (vs. hedonic).

H4b. The presence of a VI (vs. a HI) leads to a better attitude toward the brand, in particular when the type of message is utilitarian (vs. hedonic).

3. Research Method

3.1. Pretest

The pretest phase aimed at selecting the best image and evaluating the proper differentiation in terms of hedonic vs. utilitarian message of the captions to be used in the main study. We decided to use headphones as a product because it has been successfully used in past studies manipulating hedonic/utilitarian motives [66]. A total of 100 independent participants recruited through Amazon Mechanical Turk took part in a short online survey, following recent research trends [72,73]. Sensitivity analyses were conducted using G*power (version 3.1.9.7) [67], to compute the minimum effect size that could be detected given the alpha, power, and sample size [68]. For the repeated-measure *t*-test, the analysis entering two groups, alpha = 0.05, power = 0.80, one-tailed, and 100 participants indicated sufficient power to detect a small effect size, $d \geq 0.25$. For the one-way repeated ANOVA on anthropomorphism and the analysis entering one group, number of measurements = 3, correlation among measures = 0.10, alpha = 0.05, power = 0.80, and 100 participants indicated sufficient power to detect a moderate-to-small effect size, partial $\eta \geq 0.30$. For the one-way repeated ANOVA on the global evaluation of the hedonic/utilitarian message, the analysis entering one group, number of measurements = 4, correlation among measures = 0.10, alpha = 0.05, power = 0.80, and 100 participants indicated sufficient power to detect a moderate-to-small effect size, partial $\eta^2 \geq 0.25$. Effect sizes were interpreted following Cohen's guidelines [69].

First, we tested the efficacy of the captions by requesting participants to read short, written product descriptions to be used in the main study (see the Appendix for the exact wording) and to answer the two following questions: 1) To what extent do you perceive the content of the message is focused on "style and attractiveness"? 2) To what extent do you perceive the content of the message is focused on "functionality"? Items were answered by using a 7-point Likert scale (from 1 = not at all, to 7 = very much). The captions we created were rated as hedonic or utilitarian as expected (see Table 1 for descriptive statistics). The utilitarian caption was considered more focused on functionality than the hedonic caption, paired $t(99) = 3.39$, $p < 0.001$, Cohen's $d = 0.34$. Conversely, the hedonic caption was rated as focused on style and attractiveness more than the utilitarian caption, $t(99) = 2.26$, $p = 0.026$, Cohen's $d = 0.23$. Then, participants were shown three made-up Instagram posts, all presenting a real person wearing headphones, with no caption. The original pictures depicted real people, two women and a man (Fig. 1). We assessed perceived anthropomorphism, asking participants to evaluate each image on a 7-point semantic differential on the following questions: I) Looks human/Does not look human; II) Looks realistic/Does not look realistic [e.g., 70]. We also assessed the extent each picture was considered hedonic or utilitarian. First, definitions of hedonic and utilitarian were provided, then participants answered the following question: "I think the message the picture is trying to convey is", selecting on a 5-point semantic differential, from "completely utilitarian" to "completely hedonic" (Table 1). Concerning the anthropomorphism of the people in the pictures, we aimed at selecting a person rated as more neutral as possible: for the main study, the aim was to display the very same picture and present it as either a real influencer or a virtual influencer, as only consider the variation in users' evaluations that were ascribable to the disclosure of the influencer nature. Although the three characters were not different in terms of perceived human appearance, $F(2,98) = 2.66$, $p = 0.075$, partial $\eta^2 = 0.05$, one of the three pictures was rated significantly less "real" than the other two ($F(2,98) = 6.19$ $p = 0.003$, partial $\eta^2 = 0.11$, pairwise comparisons, $p \leq 0.036$), and thus was discarded. Between the two remaining pictures, we opted for the one considered more neutral in terms of global evaluation of the hedonic/utilitarian message conveyed by the image. Indeed, since we needed to add to the very same image either a hedonic or a utilitarian caption, we wanted an image as neutral as possible.

Table 1
Descriptive statistics and mean values for the background and main variables in the pretest.

Variables	Descriptives	
Age	34.72 (9.24)	
	Range: 20-65	
Gender	M = 66%	
	F = 34%	
	Other = 0%	
Frequency of social network use	5.32 (1.67)	
	Range: 1.00–7.00	
Caption's rating	Functionality	Style
Utilitarian	5.79 (1.28)	5.27 (1.59)
Hedonic	5.13 (1.60)	5.69 (1.25)
Anthropomorphism	Human	Real
Post 1	6.12 (1.06))	6.10 (1.04
Post 2	5.81 (1.32)	5.63 (1.32)
Post 3	6.00 (1.19)	5.97 (1.24)
Global utilitarian-hedonic rating of the picture		
Post 1	3.62 (1.02)	
Post 2	3.68 (1.01)	
Post 3	3.46 (1.02)	
Brand familiarity		
Beats by dr. Dre	4.66 (1.84)	
Crystalline	3.91 (2.00)	
Sony	5.70 (1.28)	
Echoff	4.37 (1.98)	

Note. Means, standard deviations, and ranges are reported. Only for gender, frequency (in %) are reported.

To avoid the potential impact of pre-existing attitudes and familiarity with existing brands, consistent with prior research we employed a fictitious brand [71]. In the pretest we checked that the selected fake brand name was not known, asking participants to rate the familiarity of four brand names, on a 7-point Likert scale (from 1 = not at all, to 7 = very much). The fictitious brand we created ("Crystalline") was rated less familiar than the other real ("Sony", and "Beats by Dr. Dre") and fictitious ("Echoff") brands presented, $F(3,97) = 18.55$, $p < 0.001$, partial $\eta^2 = 0.36$, (post hoc comparisons all ps ≤ 0.005), supporting its use (Table 1).

3.2. Participants

Participants were recruited via Amazon Mechanical Turk and took part in this study in return for monetary compensation, in line with recent research, [e.g., 72,73]. Sensitivity analyses were conducted using G*power (version 3.1.9.7) [67] to compute the minimum effect size that could be detected given the alpha, power, and sample size [68]. For the one-way MANOVA (two levels of the independent variable: virtual v. human influencer) on reactions to the influencer (3 dependent variables) the analysis entering two groups, one predictor, three response variables, alpha = 0.05, power = 0.80, and 238 participants indicated we had sufficient power to trust Pillai's $V \geq 0.044$. For the one-way MANOVA on reactions to the brand (2 dependent variables), the analysis entering two groups, one predictor, two response variables, alpha = 0.05, power = 0.80, and 238 participants indicated we had sufficient

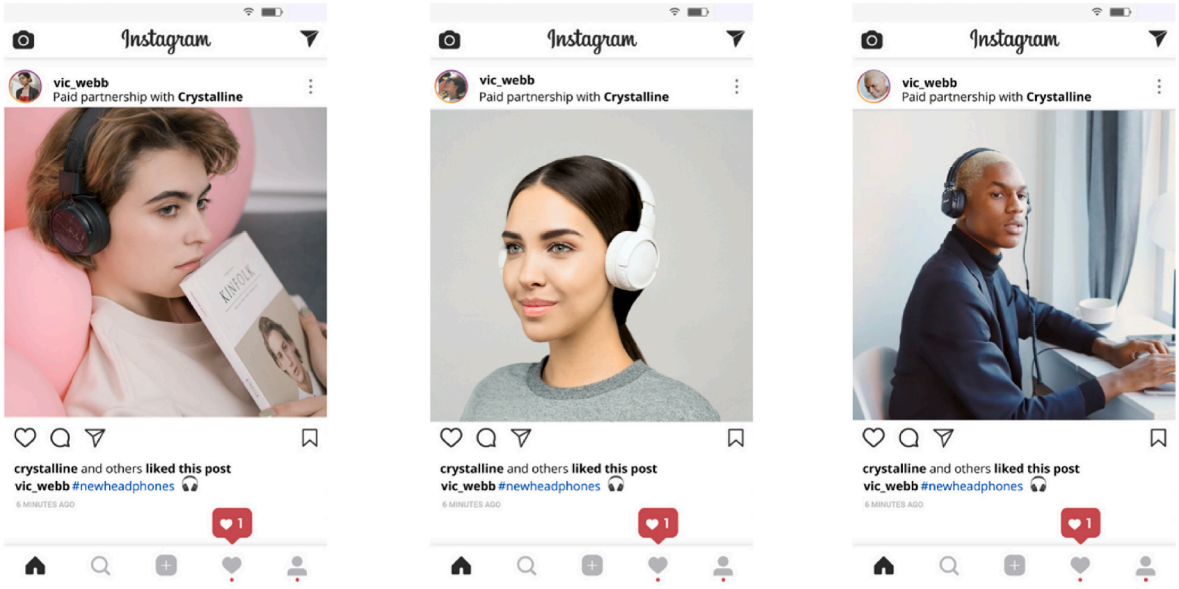


Fig. 1. The fictitious Instagram posts used in the pretest.

power to detect Pillai's V $\geq$ 0.039. For the 2 $\times$ 2 MANOVA on reactions to the influencer the analysis entering four groups, two predictors, three response variables, alpha = 0.05, power = 0.80, and 238 participants indicated we had sufficient power to trust Pillai's V $\geq$ 0.056. Finally, for the 2 $\times$ 2 MANOVA on reactions to the brand, the analysis entering four groups (two predictors and two response variables), alpha = 0.05, power = 0.80, indicated we had sufficient power to detect Pillai's V $\geq$ 0.049.

Participants who did not correctly stick to the instructions (e.g., not completing the survey; failing the attention, and checks) were excluded, which left us with a total of 238 participants for data analyses (mean age = 34.16 years $\pm$ 8.86 SD; 39% Female). Each participant provided informed consent and agreed to take part in the study. The research complies with the Declaration of Helsinki and was approved by the local Ethics Committee.

3.3. Procedure and stimuli

To select the image and the caption used in the main study, we conducted a pretest phase on an independent sample, as described in

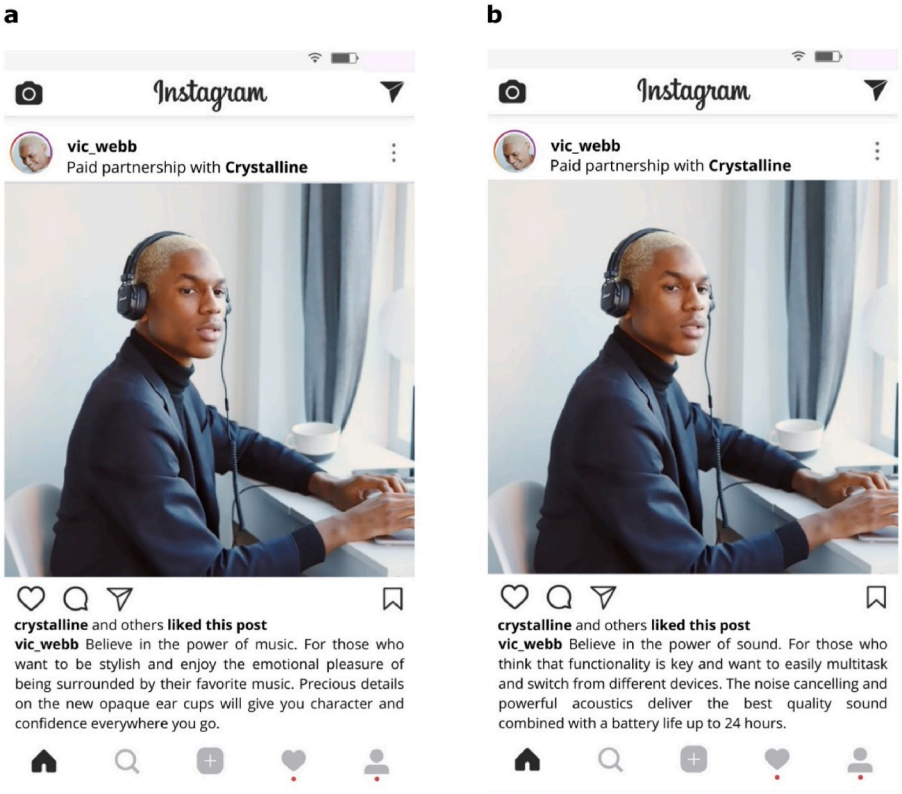


Fig. 2. The fictitious Instagram posts used in the main study.

section 3.1. Based on the results, we ended up with the post in Fig. 2.

The main study applied two factors between-subjects design (nature of the influencer: virtual vs. human; messages conveyed: hedonic vs. utilitarian). Participants were randomly assigned to one of the four possible conditions.

First, participants answered some demographic questions, and their interest in buying headphones was also assessed. Then, despite conditions, all participants were told they would be exposed to an Instagram post and were presented with the following instructions: '*Imagine scrolling your Instagram Explore Tab and finding the following post from an Influencer showcasing a new pair of headphones. Please look at the picture and read the post carefully, as we will ask you some questions about it in the following steps*'. Participants in the hedonic message condition were presented with the post with a caption focusing on appearance and emotions, while people in the utilitarian message condition saw a caption focusing on effectiveness and technical details (see the Appendix). An attention check followed, asking participants to select among six features those that were just presented in the caption (e.g., 'noise-canceling function'; 'precious details').

After that, half of the participants (human condition) were presented with the following sentence: *'The character you saw in the post promoting the headphones is an INFLUENCER'*, while for the other half (VI condition), the world *INFLUENCER* was substituted with *VIRTUAL INFLUENCER*. Then, based on the belonging condition, participants saw a brief description of what an influencer/VI is respectively. Table 2 reports both definitions provided.

In the following phase, participants reported their perceptions toward the influencer and toward the brand.

3.4. Measures

The survey included demographic questions on participants' age, gender, years of education, and social media usage, as well as measures related to the constructs of interests, which were introduced with the question 'How do you feel about the influencer/brand in the post?' and measured through 7-point differential semantic scales.

Social presence was measured following [74] and included 6 items (e.g. Cold/Warm).

Trust toward the influencer was measured following [75] and included 11 items (e.g. Unreliable/Reliable).

Attitude toward the influencer was measured following [76] and included 6 items (e.g. Unappealing/Appealing).

Trust toward the brand was measured following [75] and included 11 items (e.g. Untrustworthy/Trustworthy).

Attitude toward the brand was measured following [76] and included 6 items (e.g. Bad/Good).

Information about the items' loadings and reliability is displayed in Table 3, while the full list of items is reported in the Appendix.

Table 2
Influencer and virtual influencer's definitions provided in the study.

Definition of "influencer"	Definition of "virtual influencer"
Influencers are regular social media (e.g., Instagram) users but with a unique ability to influence others, thanks to a well-established credibility and a significant audience. From a marketing perspective, an influencer is defined as a person with the power to influence potential buyers of a product or service, because of their authority, knowledge, position, or relationship with their audience. Increasingly often, influencers are paid by brands to promote their products, generally within a specific niche.	Virtual influencers are computer-generated influencers. Both their digital character and personality are completely artificial: the former is created in computer graphics; the latter is defined through a purposely scripted narrative. As for human influencers, they are active on social media such as Instagram, often featuring brand marketing campaigns to promote services and products. Thanks to advancements in AI research and technology, virtual influencers have realistic physical appearances and human-like personalities.

Table 3
Items' loadings and reliability.

	Items	Mean (SD)	Loadings	Cronbch's Alpha	CR	AVE
Social presence				0.876	0.876	0.638
	SP1	5.58 (1.21)	0.638			
	SP2	5.34 (1.32)	0.791			
	SP3	5.66 (1.22)	0.774			
	SP4	5.43 (1.33)	0.754			
	SP5	5.66 (1.22)	0.727			
	SP6	5.21 (1.44)	0.754			
Attitude twd the influencer				0.896	0.896	0.591
	ATT1	5.53 (1.28)	0.760			
	ATT2	5.53 (1.21)	0.789			
	ATT3	5.61 (1.33)	0.709			
	ATT4	5.59 (1.22)	0.763			
	ATT5	5.68 (1.22)	0.805			
	ATT6	5.64 (1.24)	0.784			
Trust twd the influencer				0.939	0.940	0.595
	TR1	5.43 (1.44)	0.753			
	TR2	5.30 (1.39)	0.797			
	TR3	5.34 (1.31)	0.837			
	TR4	5.35 (1.34)	0.832			
	TR5	5.39 (1.35)	0.854			
	TR6	5.44 (1.30)	0.740			
	TR7	5.52 (1.24)	0.764			
	TR8	5.47 (1.30)	0.768			
	TR9	5.55 (1.18)	0.762			
	TR10	4.97 (1.50)	0.559			
	TR11	5.40 (1.30)	0.780			
Attitude twd the brand				0.913	0.913	0.638
	ATT1	5.54 (1.29)	0.725			
	ATT2	5.48 (1.32)	0.789			
	ATT3	5.60 (1.28)	0.800			
	ATT4	5.55 (1.26)	0.800			
	ATT5	5.61 (1.31)	0.862			
	ATT6	5.58 (1.34)	0.809			
Trust twd the brand				0.941	0.942	0.603
	TR1	5.58 (1.20)	0.787			
	TR2	5.60 (1.22)	0799			
	TR3	5.57 (1.29)	0.812			
	TR4	5.50 (1.18)	0.758			
	TR5	5.61 (1.26)	0.844			

(*continued on next page*)

Table 3 (*continued*)

	Items	Mean (SD)	Loadings	Cronbch's Alpha	CR	AVE
	TR6	5.45 (1.35)	0.849			
	TR7	5.47 (1.29)	0.798			
	TR8	5.51 (1.19)	0.739			
	TR9	5.56 (1.20)	0.720			
	TR10	5.06 (1.43)	0.591			
	TR11	5.51 (1.25)	0.812			

4. Results

4.1. Preliminary analyses

First, we verified that participants in the four experimental conditions did not differ in background variables. Gender distribution was balanced in the four conditions, χ^2 (6) = 4.70, p = 0.583. Furthermore, two-way ANOVAs with Influencer (human vs. virtual) and Message (hedonic vs. utilitarian) entered as independent factors indicated no significant differences between the four groups in age, F (1, 234) ≤ 1.19, p ≥ 0.277, partial $\eta^2 \leq 0.005$, education, F(1, 224) ≤ 2.55, p ≥ 0.112, partial $\eta^2 \leq 0.011$, and frequency of SNS use, F(1, 234) ≤ 2.28, p ≥ 0.132, partial $\eta^2 \leq 0.010$.

Table 4 reports all the descriptive statistics for the background variables in the four experimental conditions.

4.2. Effects of the nature of the influencer on social presence, trust, and attitude toward the influencer

First, we examined whether the nature of the Influencer (human vs. virtual) affected people's reactions to the influencer in terms of social presence, trust, and attitude. We performed a MANOVA entering Influencer nature as an independent variable and attitude, trust, and social presence toward the influencer as dependent variables. Multivariate results indicated that the nature of the Influencer (human vs. virtual), V = 0.005, F(3, 234) = 0.37, p = 0.773, partial $\eta^2 = 0.005$, did not result in any statistically significant difference in the users' perceptions of social presence, trust, and attitude toward the influencer, thus respectively confirming H1a, H1b, and H1c.

Table 4
Descriptive statistics for the background variables in the four experimental conditions.

	Human Influencer		Virtual Influencer	
	Hedonic Message (*n* = 53)	**Utilitarian Message** (*n* = 66)	**Hedonic Message** (*n* = 56)	**Utilitarian Message** (*n* = 63)
Age	34.89 (10.24) 21–70	33.52 (8.27) 19–67	33.54 (7.33) 23–53	34.76 (10.43) 19–66
Gender	M = 66%, F = 34% Other = 0%	M = 56%, F = 44% Other = 0%	M = 57%, F = 43% Other = 0%	M = 63%, F = 35% Other = 2%
Education	15.53 (5.12) 2–24	13.38 (5.33) 1–21	14.55 (4.81) 3–30	14.62 (5.56) 1–30
Frequency of social network use	5.37 (1.11) 1.80–6.80	5.01 (1.18) 1.80–7.00	5.35 (1.10) 2.40–7.00	5.25 (1.14) 2.20–7.00

Note. Means, standard deviations, and ranges are reported. Only for gender, the frequency (in %) is reported.

4.3. Effects of the nature of the influencer on trust and attitude toward the brand

A MANOVA was conducted entering Influencer nature (human vs. virtual) as independent variable and attitude and trust toward the brand as dependent variables. Multivariate results indicated that the nature of the Influencer (human vs. virtual), V = 0.001, F(2, 235) = 0.16, p = 0.852, partial $\eta^2 = 0.001$, did not result in any statistically significant difference in the users' perceptions of trust and attitude toward the brand, thus supporting H2a and H2b.

Table 5 presents means and standard deviations in the four experimental conditions.

4.4. Interactive effects of nature of the influencer and type of message conveyed on influencer and brand outcomes

A 2 × 2 MANOVA was conducted on trust and attitude toward the influencer with the nature of the Influencer (human vs. virtual) and type of message (hedonic vs. utilitarian) as independent variables. Contrary to H3a and H3b, the interaction effect was not significant, V = 0.018, F (3, 232) = 1.42, p = 0.239, partial $\eta^2 = 0.018$. Thus, the presence of a virtual (vs. human) influencer does not affect trust and attitude toward the influencer when the type of message is utilitarian (vs. hedonic).

A similar 2 × 2 MANOVA was conducted on trust and attitude toward the brand, with the nature of the Influencer (human vs. virtual) and type of message (hedonic vs. utilitarian) as independent variables. Also in this case, contrary to H4a and H4b, the interaction effect was not significant, V = 0.020, F(2, 233) = 2.40, p = 0.093, partial $\eta^2 = 0.020$. Thus, the presence of a virtual (vs. human) influencer does not affect trust and attitude toward the brand when the type of message is utilitarian (vs. hedonic).

Table 6 summarizes all the results of the hypotheses testing.

5. Discussion

Addressing the recent call for research investigating the relation between VIs and human endorsers from the consumer's perspective [10,15,77] this study applied an experimental design to explicate how consumers react to human-like VIs compared to HIs. Furthermore, we examined the conditions – specifically, the type of message communicated (hedonic vs. utilitarian) - under which using a virtual influencer rather than a human influencer can optimize their potential in terms of trust and attitude both toward the influencer and the brand. Our research focused on recent virtual influencer literature, aiming to advance and test hypotheses grounded in solid theory. To create a comprehensive study, we explored various hypotheses, including those predicting no significant differences in users' perceptions between AI and human influencers. This approach aligns with McShane et al. [78], emphasizing transparent result presentation for generalized conclusions based on cumulative evidence. Our goal was to provide insightful results, acting as a benchmark for past and future research comparisons.

Our experimental study showed that the nature of the influencer – human vs. virtual – does not significantly affect users' social presence, trust, and attitude toward the influencer (hypotheses 1a, 1b, and 1c supported). Our results side with Stein and colleagues [31], who did not observe any advantage of a human over a virtual influencer in terms of parasocial relationships. As reported by Liu [79], this might be due to people's need to reduce uncertainty about a non-human agent which motivates them to see the humanness in it and experience the same level of social presence perceived with human endorsers. Present results also align with Sands and colleagues [20], who found similarities in how people respond to AI and human influencers, and with Yam and colleagues [80], arguing that VIs can be treated as social beings by their followers if they are fully anthropomorphized or human-like. The main insight here is that the results fit well with the CASA paradigm, suggesting that users respond to technologies by employing social rules they

Table 5
Means and standard deviations for perceptions toward the influencer and toward the brand in the four experimental conditions.

	Human Influencer		Virtual Influencer	
	Hedonic Message ($n = 53$)	**Utilitarian Message** ($n = 66$)	**Hedonic Message** ($n = 56$)	**Utilitarian Message** ($n = 63$)
Influencer				
Attitude	5.74 (1.04)	5.43 (1.06)	5.51 (0.96)	5.72 (0.97)
Trust	5.64 (0.97)	5.19 (1.15)	5.34 (0.90)	5.39 (1.11)
Social Presence	5.65 (0.98)	5.25 (1.07)	5.42 (0.88)	5.42 (1.17)
Brand				
Attitude	5.72 (1.13)	5.49 (1.16)	5.45 (0.87)	5.60 (1.14)
Trust	5.67 (0.99)	5.38 (1.13)	5.36 (0.90)	5.58 (0.93)

Table 6
Summary of the hypotheses' testing results.

Hypothesis	Manipulation	Dependent Variable	Result
H1a	Type of Influencer (VI vs HI)	Social Presence toward the Influencer	No statistically significant difference. Hypothesis supported.
H1b		Trust toward the Influencer	No statistically significant difference. Hypothesis supported.
H1c		Attitude toward the Influencer	No statistically significant difference. Hypothesis supported.
H2a		Trust toward the Brand	No statistically significant difference. Hypothesis supported.
H2b		Attitude toward the Brand	No statistically significant difference. Hypothesis supported.
H3a	Type of Influencer (VI vs HI) x Type of Message (Utilitarian vs Hedonic)	Trust toward the Influencer	No statistically significant interaction effect. Hypothesis not supported.
H3b		Attitude toward the Influencer	No statistically significant interaction effect. Hypothesis not supported.
H4a		Trust toward the Brand	No statistically significant interaction effect. Hypothesis not supported.
H4b		Attitude toward the Brand	No statistically significant interaction effect. Hypothesis not supported.

already applied to human-human interactions [81]. Our findings support the view that human beings tend to anthropomorphize machines and abide by social rules in their interactions with them [21,36,82]. Given that VIs tend to have human-like, attractive appearances and exhibit unique lifestyles and personalities, users are likely to perceive these influencers as genuine social actors when interacting, at least until the virtual influencer incorporates excessive reality into a single social media post, like consuming a branded product alongside a real human [83]. On the opposite, the current findings do not support the results of studies suggesting that people - all other things being equal - tend to prefer humans over digital agents [94] and VIs [7]. The disclosure of the influencer's nature did not trigger a sense of eeriness and unease in the users [84]. This outcome aligns partially with Li and colleagues [61], who found virtual influencers to be less effective than HIs only when exhibiting synthetic authenticity through artificial cues. Rather, when sensory cue salience was low, that is when the appearance of the virtual influencer closely resembles that of a human without displaying synthetic artifacts - as observed in our study - VIs were comparable to HIs (see also [85]). We speculated that the absence of differences in social presence, trust, and attitude toward the influencer in our comparison between a human and a virtual influencer can be attributed to the heightened significance placed on the visual and aesthetic appearance over the disclosure of the artificial nature of the influencer. That is, users are more likely to focus on the aesthetic appeal presented in the Instagram post, diminishing the impact of any artificial disclosure on users' perceptions of warmth, friendliness, emotional capacity (core aspects of social presence), reliability, credibility, and conclusiveness (core aspects of trust), and on users' general attitude toward the influencer.

The absence of substantial differences in attitude, social presence, and trust offers notable benefits for businesses adopting virtual influencers. However, it must be noted that this scenario may introduce "inherent tensions" within relational dynamics, and further research is unquestionably required to explore whether engaging with these virtual entities might result in potential dehumanization and propensity towards a diminishing of human relationships [86].

Furthermore, our results indicate that using an AI human-like influencer instead of a real human one does not entail any significant differences in terms of brand perceptions, supporting hypotheses 2a and 2b. These results align with Thomas and Fowler's findings [15] where no significant differences between a virtual influencer and a celebrity endorser were found in terms of attitude toward the brand and purchase intention. The result is of particular interest to companies since whatever the adopted solution (virtual or human), the outcome should not differ. In terms of brand trust, our study lines up with the findings of Muniz and colleagues [87], where disclosure of the human vs virtual nature of the influencer does not directly result in differences in brand perceptions for users. To be noted, our results diverge from Li et al. [61], who found that human influencers are more effective in influencing brand attitudes (and purchase intention) compared to virtual influencers. These conflicting results regarding brand-related outcomes suggest the involvement of alternative mechanisms when users interact with virtual influencers, warranting further studies to disentangle this phenomenon.

Finally, our study explored whether a virtual influencer can effectively leverage emotions to promote a brand's product, despite its non-human nature and the lack of sensory capability. In this regard, we did not find evidence supporting the relevance of the type of message conveyed by the influencer - hedonic vs. utilitarian - and the influencer's nature (all hypotheses 3 and 4 were not supported). Although this finding might be partially explained considering that users are not always motivated to process AI's messages carefully, these results deserve special attention as they further indicate that individuals may not hold

fundamentally different schemas about human-like VIs.

Recent findings suggest the importance of striking a balance in attributing purely human cues to a virtual influencer, as simply simulating typically human behaviors may prove counterproductive [83]. In our study, manipulating the message with a more hedonic or instrumental tone does not seem to reflect a different sensory capability resulting in a difference in the influencer and brand perceptions. This is in contrast to Li et al. [61], where the endorsement effectiveness of virtual influencers can be enhanced when sensory cue salience is low.

Our results are particularly interesting in light of the brands' urgency to satisfy customers' needs and wants by leveraging proper messages and effective communication in promotion strategies [88]. Rather than a thinking entity able to decide, process, and create information autonomously, our findings suggest that VI are actually perceived as advocates for a message that humans (e.g. communication teams or brands themselves) have priorly established, thus mitigating the impact of AI aversion and generating a more neutral reception and acceptance. The awareness that VIs are not acting autonomously avoids the rise of concerns regarding moral or ethical judgment [89] and could further explain the lack of differences in VI perception and brand outcomes as a function of the type of conveyed message.

6. Theoretical and managerial implications

Our research makes important theoretical contributions and practical implications. First, the study constitutes a valuable addition to the existing literature on the analysis of virtual influencers [5,24,83]. Specifically, our findings contribute to the literature on human-virtual influencer interactions, by advancing the debate on whether users engage with humanlike artificial entities in the same way as with other humans [31]. Also, the study addresses the ongoing controversy on the use and effectiveness of AI in influencer marketing [90], as well as Huang and Rust's [91] question about how marketers and AI should collaborate to better resonate with customers' needs and preferences. Our study offers empirical support for the CASA paradigm, at the same time stimulating the discussion between prominent theories examining people's reactions to emerging technology interfaces and artificial intelligence (speciesism [77], algorithmic aversion [78], Uncanny Valley [32]). Although a more in-depth examination of the phenomenon is required, our findings provide evidence that when anthropomorphism levels are high, without any visual cues that might evoke uncanny feelings of "*almost human, but not quite*", users' evaluations are not biased from speciesism or algorithm aversion and do not develop an aversion or a hostility towards non-humans. Additionally, by hypothesizing the interplay of the type of message conveyed by the influencer and brand-related variables, we advance knowledge on the influencer-product fit literature [23,27,55], as well as research assessing whether people consider emotion and sensory feelings as human prerogatives [61].

From a practical perspective, our findings offer insights for brand marketing managers in terms of the risks and benefits of incorporating a virtual ambassador into their pool of social media-based branding activities. Brands may trust digital ambassadors over real ambassadors for a variety of reasons, such as mitigating the risk of a negative spillover effect from the endorsers' private behavior or boosting the perceived innovativeness of their brand. Notably, it appears that a completely humanlike virtual endorser can deliver the same effects as a human one, even if its AI-based nature is revealed. Accordingly, perfectly anthropomorphized VIs are not subject to the same biases and stereotypes as robots and AIs engaged in human activities, and their coexistence with human pools of endorsers can be complementary rather than contradictory within a broader ambassador campaign, even when such campaign aims at leveraging emotional-based rather than cognitive-oriented product features.

In terms of implications for policy-makers and consumer protection, our findings support the vision of thinking of VIs as a socially disruptive technology that makes the distinction between virtual and real more challenging [92]. Since users' perceptions of virtual endorsers and human celebrities are similar, even when the digital nature of the former is disclosed, regulations and clear declarations on the part of brands concerned about deceptive practices are fundamental - particularly when hyper-humanized figures are involved, in order to protect users from untruthful information.

7. Limitations and directions for future research

This research has some limitations that should be addressed in future research. First, we recruited participants via Amazon Mechanical Turk. Notwithstanding the widespread use of MTurk in research, the generalizability of the results obtained to the general population is not guaranteed [93]. Future studies should explore reactions to VIs in a more heterogeneous population. Second, our study presented a single-spot encounter with the influencer, limiting the possibility of analyzing other influencer-related features that may maximize the effectiveness of digital influencers' endorsement. To study the relational exchange in terms of trust between an influencer and a brand, and understand how one impacts the other, context-specific information, including influencers' personal history and unique characteristics, is crucial. Also, users' repeated exposure to the influencer and the brand endorsed, and a growing narrative built over time, is essential. All these specifics were omitted from the current study as they deviate from the primary goal. Since examining how trust established by the VI influences the associated brand, and vice versa could contribute to a deeper comprehension of the evolving dynamics in the digital marketing landscape, future studies should quantify possible reciprocal spillover effects between influencers and brands.

Future research should differentiate the stimuli and test different VIs' characteristics, exploring the impact of relevant variables such as authenticity [18]. As anticipated the virtual influencer portrayed in our study was indeed a human that, consequently, had high levels of anthropomorphism. Methodologically speaking, this choice was motivated by the fact that we aimed at capturing only the variation of users' reactions that was directly ascribable to the disclosure of the human/non-human nature of the endorser, eliminating the impact of potentially disturbing odd features typical of digital avatars. As virtual influencers populating our digital platforms are indeed more and more indistinguishable from human ones, we are confident about the goodness of the person portrayed in our stimuli. However, we acknowledge that further studies are needed to assess the validity of our findings across a wider range of human-like characters. Specifically, future studies should examine if and how different levels of anthropomorphism and humanness can impact users' evaluations of VIs through the activation of aversive systems through the uncanny valley [95]. Moreover, our findings exclude the possibility that users' appreciation for both VIs and HIs could be attributed to a mindless-driven unawareness of being exposed to synthetic content, because users in the VIs scenario were informed that the digital persona they were observing was indeed an AI-generated character through a disclosure. However, further studies - inspired by research in the native advertising and branded content literature [e.g., 57] - might want to assess if and how different levels of human likeness require different levels of disclosure prominence, as well as disentangle the potential effect of the presence/absence of such AI disclosure on users' perceived transparency and trust.

This latter point raises a further interesting issue related to body image portrayed on Instagram - and social media in general [38]. In our study, we adopted a stimulus that features a human that, however, is adherent to mainstream Instagram beauty standards (e.g. perfect skin, slender body shape). At first instance, probably the skin retouching and sculpting in Photoshop, or other filtering effects made the fictitious influencer a truly suitable option to ensure the effectiveness of our disclosure-based manipulation in both the VI and human condition. At the same time, as reported by Lou et al. [18], the VIs' "curated flaws"

might have contributed to a higher level of acceptance, eliminating the eeriness that might arise from extremely perfect, doll-like individuals. Future research could delve into the issue of VIs' perceived authenticity when compared with extremely retouched and staged pictures of HIs.

Finally, in our study, we presented the influencers in a professional setting conveying professionalism and reliability, as we believed that this minimized the potential race/gender biases in users' perceptions. However, following studies might want to deepen our understanding of the VIs phenomenon also from a race and gender perspective, analyzing the impact of stereotypes and biases, as well as crossing such themes with the topic of body diversity and acceptance.

Conflicts of interest

The authors declare that the research was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest.

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Research transparency statement

The authors reported how sample size was determined, all data exclusions, all manipulations, and all measures in the studies.

CRediT authorship contribution statement

Roberta De Cicco: Conceptualization, Investigation, Project administration, Supervision, Writing – original draft, Writing – review & editing. **Serena Iacobucci:** Conceptualization, Writing – original draft. **Loreta Cannito:** Data curation, Methodology. **Gianni Onesti:** Project administration. **Irene Ceccato:** Funding acquisition, Methodology, Resources, Software. **Riccardo Palumbo:** Funding acquisition, Supervision.

Data availability

Data will be made available on request.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.techsoc.2024.102488.

References

[1] E.-J. Lee, Authenticity model of (mass-oriented) computer-mediated communication: Conceptual explorations and testable propositions, J. Comput. Commun. 25 (2020) 60–73, https://doi.org/10.1093/jcmc/zmz025.
[2] Miquela Net Worth NetWorthSpot, Earnings. https://www.networthspot.com/lilmiquela/net-worth/instagram/, 2020. (Accessed 18 January 2022).
[3] L. Dahlberg, Computer-mediated communication and the public sphere: a critical analysis, J. Comput. Commun. 7 (2006), https://doi.org/10.1111/j.1083-6101.2001.tb00137.x.
[4] L.V. Casaló, C. Flavián, S. Ibáñez-Sánchez, Influencers on Instagram: Antecedents and consequences of opinion leadership, J. Bus. Res. 117 (2020) 510–519, https://doi.org/10.1016/j.jbusres.2018.07.005.
[5] L. Xie-Carson, T. Magor, P. Benckendorff, K. Hughes, All hype or the real deal? Investigating user engagement with virtual influencers in tourism, Tourism Manag. 99 (2023) 104779, https://doi.org/10.1016/j.tourman.2023.104779.
[6] K. Xu, T. Liao, Explicating cues: a typology for understanding emerging media technologies, J. Comput. Commun. 25 (2020) 32–43, https://doi.org/10.1093/JCMC/ZMZ023.
[7] J. Arsenyan, A. Mirowska, Almost human? A comparative case study on the social media presence of virtual influencers, Int. J. Hum. Comput. Stud. 155 (2021) 102694, https://doi.org/10.1016/j.ijhcs.2021.102694.
[8] Virtual Influencers Speak Out in Response to the Murder of George Floyd, VirtualHumans.org, 2020. https://www.virtualhumans.org/article/virtual-influencers-speak-out-in-response-to-the-murder-of-george-floyd. (Accessed 18 January 2022).
[9] Ga.Agency, What Are Virtual Influencers and How Can They Help Your Future Brand Awareness?, 2020. https://www.ga.agency/en/blog/what-are-virtual-influencers-and-what-they-mean-for-your-brand. (Accessed 18 January 2022).
[10] A. Mirowska, J. Arsenyan, Sweet escape: the role of empathy in social media engagement with human versus virtual influencers, Int. J. Hum. Comput. Stud. 174 (2023) 103008, https://doi.org/10.1016/j.ijhcs.2023.103008.
[11] Virtualhumans.org, Reasons Brands Love to Work with Virtual Influencers, 2021, p. 4. https://www.virtualhumans.org/. (Accessed 18 January 2022).
[12] R. Schwarz, Should Marketers Leverage Virtual Influencers? Forbes, 2022. https://www.forbes.com/sites/forbescommunicationscouncil/2022/05/23/should-marketers-leverage-virtual-influencers/?sh=2b41c2e44b6a. (Accessed 17 September 2023).
[13] S. Khamis, L. Ang, R. Welling, Self-branding, 'micro-celebrity' and the rise of social media influencers, Celebr. Stud. 8 (2017) 191–208, https://doi.org/10.1080/19392397.2016.1218292.
[14] H. Reinikainen, T.M. Tan, V. Luoma-aho, J. Salo, Making and breaking relationships on social media: the impacts of brand and influencer betrayals, Technol. Forecast. Soc. Change 171 (2021) 120990, https://doi.org/10.1016/j.techfore.2021.120990.
[15] V.L. Thomas, K. Fowler, Close encounters of the AI kind: use of AI influencers as brand endorsers, J. Advert. 50 (2021) 11–25, https://doi.org/10.1080/00913367.2020.1810595.
[16] E.C.-X. Aw, S.H.-W. Chuah, "Stop the unattainable ideal for an ordinary me!" fostering parasocial relationships with social media influencers: the role of self-discrepancy, J. Bus. Res. 132 (2021) 146–157, https://doi.org/10.1016/j.jbusres.2021.04.025.
[17] M.D. Hanus, J. Fox, Persuasive avatars: the effects of customizing a virtual salesperson's appearance on brand liking and purchase intentions, Int. J. Hum. Comput. Stud. 84 (2015) 33–40, https://doi.org/10.1016/j.ijhcs.2015.07.004.
[18] C. Lou, S.T.J. Kiew, T. Chen, T.Y.M. Lee, J.E.C. Ong, Z. Phua, Authentically fake? How consumers respond to the influence of Virtual Influencers, J. Advert. 52 (2023) 540–557, https://doi.org/10.1080/00913367.2022.2149641.
[19] S. Sands, C. Ferraro, C. Campbell, H.-Y. Tsao, Managing the human–chatbot divide: how service scripts influence service experience, J. Serv. Manag. 32 (2021) 246–264, https://doi.org/10.1108/JOSM-06-2019-0203.
[20] S. Sands, C.L. Campbell, K. Plangger, C. Ferraro, Unreal influence: leveraging AI in influencer marketing, Eur. J. Market. 56 (2022) 1721–1747, https://doi.org/10.1108/EJM-12-2019-0949.
[21] J. Feine, U. Gnewuch, S. Morana, A. Maedche, A Taxonomy of social cues for conversational agents, Int. J. Hum. Comput. Stud. 132 (2019) 138–161, https://doi.org/10.1016/j.ijhcs.2019.07.009.
[22] M.A. Selamat, N.A. Windasari, Chatbot for SMEs: Integrating customer and business owner perspectives, Technol. Soc. 66 (2021) 101685, https://doi.org/10.1016/J.TECHSOC.2021.101685.
[23] H. Kim, M. Park, Virtual influencers' attractiveness effect on purchase intention: a moderated mediation model of the Product–Endorser fit with the brand, Comput. Hum. Behav. 143 (2023) 107703, https://doi.org/10.1016/j.chb.2023.107703.
[24] R.J. Ahn, S.Y. Cho, W. Sunny Tsai, Demystifying Computer-Generated Imagery (CGI) Influencers: the effect of perceived anthropomorphism and social presence on brand outcomes, J. Interact. Advert. 22 (2022) 327–335, https://doi.org/10.1080/15252019.2022.2111242.
[25] J. Sheena, Brands are still figuring out virtual influencers, Mark. Brew. (2023). https://www.marketingbrew.com/stories/2023/09/12/brands-are-still-figuring-out-virtual-influencers. (Accessed 21 December 2023).
[26] H.P. Bowen, M.F. Wiersema, Matching method to paradigm in strategy research: limitations of cross-sectional analysis and some methodological alternatives, Strat. Manag. J. 20 (1999) 625–636, https://doi.org/10.1002/(SICI)1097-0266, 199907)20:7<625::AID-SMJ45>3.0.CO;2-V.
[27] L. Janssen, A.P. Schouten, E.A.J. Croes, Influencer advertising on Instagram: product-influencer fit and number of followers affect advertising outcomes and influencer evaluations via credibility and identification, Int. J. Advert. 41 (2022) 101–127, https://doi.org/10.1080/02650487.2021.1994205.
[28] R.F. Lusch, S.L. Vargo, Service-dominant logic — a guiding framework for inbound marketing, Mark. Rev. St. Gall. 26 (2009) 6–10, https://doi.org/10.1007/s11621-009-0094-6.
[29] B. Vlačić, L. Corbo, S. Costa e Silva, M. Dabić, The evolving role of artificial intelligence in marketing: a review and research agenda, J. Bus. Res. 128 (2021) 187–203, https://doi.org/10.1016/j.jbusres.2021.01.055.
[30] K.J. Byun, S.J. Grace Ahn, A systematic review of virtual influencers: similarities and differences between human and virtual influencers in interactive advertising, J. Interact. Advert. 23 (2023) 293–306, https://doi.org/10.1080/15252019.2023.2236102.
[31] J.P. Stein, P. Linda Breves, N. Anders, Parasocial Interactions with Real and Virtual Influencers: the Role of Perceived Similarity and Human-Likeness, New Media Soc, 2022, https://doi.org/10.1177/14614448221102900.
[32] H. Kim, M. Park, When digital celebrity talks to you: how human-like virtual influencers satisfy consumer's experience through social presence on social media endorsements, J. Retailing Consum. Serv. 76 (2024) 103581, https://doi.org/10.1016/j.jretconser.2023.103581.

[33] S.V. Jin, A. Muqaddam, E. Ryu, Instafamous and social media influencer marketing, Market. Intell. Plann. 37 (2019) 567–579, https://doi.org/10.1108/MIP-09-2018-0375.
[34] B.R. Gaines, From facilitating interactivity to managing hyperconnectivity: 50 years of human–computer studies, Int. J. Hum. Comput. Stud. 131 (2019) 4–22, https://doi.org/10.1016/j.ijhcs.2019.05.007.
[35] Y. Kim, S.S. Sundar, Anthropomorphism of computers: is it mindful or mindless? Comput. Hum. Behav. 28 (2012) 241–250, https://doi.org/10.1016/J.CHB.2011.09.006.
[36] C. Nass, Y. Moon, Machines and mindlessness: social responses to computers, J. Soc. Issues 56 (2000) 81–103, https://doi.org/10.1111/0022-4537.00153.
[37] M. Lombard, K. Xu, Social responses to media technologies in the 21st century: the media are social actors paradigm, Human-Machine Commun 2 (2021) 29–55, https://doi.org/10.30658/hmc.2.2.
[38] C. Chen, O. Cohen, S.S. Sundar, Differentiating problematic from habitual Instagram use: a uses and grats 2.0 perspective, Soc. Media + Soc. 8 (2022) 205630512211163, https://doi.org/10.1177/20563051221116339.
[39] M. Mori, Bukimi no tani [the uncanny valley], Energy 7 (1970) 33–35.
[40] M. Mori, K.F. MacDorman, N. Kageki, The uncanny valley, IEEE Robot. Autom. Mag. 19 (2012) 98–100, https://doi.org/10.1109/MRA.2012.2192811.
[41] C. Baumgarth, A. Kirkby, C. Kaibel, When fake becomes real: the innovative case of artificial influencers, in: E. Pantano (Ed.), Creat. Mark. Fuel Success, Emerald Publishing Limited, 2021, pp. 149–167, https://doi.org/10.1108/978-1-80071-330-720211010.
[42] Straub Gefen, Managing user trust in B2C E-services, E-Service J. 2 (2003) 7, https://doi.org/10.2979/esj.2003.2.2.7.
[43] J. Kim, K. Merrill Jr., C. Collins, H. Yang, Social TV viewing during the COVID-19 lockdown: the mediating role of social presence, Technol. Soc. 67 (2021) 101733, https://doi.org/10.1016/j.techsoc.2021.101733.
[44] K. Hassanein, M. Head, Manipulating perceived social presence through the web interface and its impact on attitude towards online shopping, Int. J. Hum. Comput. Stud. 65 (2007) 689–708, https://doi.org/10.1016/j.ijhcs.2006.11.018.
[45] C. Jiang, R.M. Rashid, J. Wang, Investigating the role of social presence dimensions and information support on consumers' trust and shopping intentions, J. Retailing Consum. Serv. 51 (2019) 263–270, https://doi.org/10.1016/j.jretconser.2019.06.007.
[46] N. Ameen, A. Tarhini, A. Reppel, A. Anand, Customer experiences in the age of artificial intelligence, Comput. Hum. Behav. 114 (2021) 106548, https://doi.org/10.1016/j.chb.2020.106548.
[47] I. Ajzen, in: L. Berkowitz (Ed.), Attitudes, Traits, and Actions: Dispositional Prediction of Behavior in Personality and Social Psychology, Adv. Exp. Soc. Psychol., Academic Press, 1987, pp. 1–63, https://doi.org/10.1016/S0065-2601(08)60411-6.
[48] A.P. Schouten, L. Janssen, M. Verspaget, Celebrity vs. Influencer endorsements in advertising: the role of identification, credibility, and Product-Endorser fit, Int. J. Advert. 39 (2020) 258–281, https://doi.org/10.1080/02650487.2019.1634898.
[49] D. Vrontis, A. Makrides, M. Christofi, A. Thrassou, Social media influencer marketing: a systematic review, integrative framework and future research agenda, Int. J. Consum. Stud. 45 (2021) 617–644, https://doi.org/10.1111/IJCS.12647.
[50] R.B. Cialdini, Influence: the Psychology of Persuasion, Morrow, New York, 1993.
[51] Y. Shan, K.-J. Chen, J-S, Elaine, Lin, When social media influencers endorse brands: the effects of self-influencer congruence, parasocial identification, and perceived endorser motive, Int. J. Advert. 39 (2020) 590–610, https://doi.org/10.1080/02650487.2019.1678322.
[52] G. McCracken, Who is the celebrity endorser? Cultural foundations of the endorsement process, J. Consum. Res. 16 (1989) 310, https://doi.org/10.1086/209217.
[53] P. Torres, M. Augusto, M. Matos, Antecedents and outcomes of digital influencer endorsement: an exploratory study, Psychol. Market. 36 (2019) 1267–1276, https://doi.org/10.1002/mar.21274.
[54] M. Delbaere, B. Michael, B.J. Phillips, Social media influencers: a route to brand engagement for their followers, Psychol. Market. 38 (2021) 101–112, https://doi.org/10.1002/mar.21419.
[55] L. Bergkvist, H. Hjalmarson, A.W. Mägi, A new model of how celebrity endorsements work: attitude toward the endorsement as a mediator of celebrity source and endorsement effects, Int. J. Advert. 35 (2016) 171–184, https://doi.org/10.1080/02650487.2015.1024384.
[56] F.F. Leung, F.F. Gu, R.W. Palmatier, Online influencer marketing, J. Acad. Market. Sci. 50 (2022) 226–251, https://doi.org/10.1007/s11747-021-00829-4.
[57] R. De Cicco, S. Iacobucci, S. Pagliaro, The effect of influencer–product fit on advertising recognition and the role of an enhanced disclosure in increasing sponsorship transparency, Int. J. Advert. 40 (2021) 733–759, https://doi.org/10.1080/02650487.2020.1801198.
[58] F. Zhou, Q. Su, J. Mou, Understanding the effect of website logos as animated spokescharacters on the advertising: a lens of parasocial interaction relationship, Technol. Soc. 65 (2021) 101571, https://doi.org/10.1016/j.techsoc.2021.101571.
[59] H.-C. Lin, P.F. Bruning, H. Swarna, Using online opinion leaders to promote the hedonic and utilitarian value of products and services, Bus. Horiz. 61 (2018) 431–442, https://doi.org/10.1016/j.bushor.2018.01.010.
[60] H. Shoenberger, E. Anna, Kim, Y. Sun, Advertising during COVID-19: exploring perceived brand message authenticity and potential psychological reactance, J. Advert. 50 (2021) 253–261, https://doi.org/10.1080/00913367.2021.1927914.
[61] H. Li, Y. Lei, Q. Zhou, H. Yuan, Can you sense without being human? Comparing virtual and human influencers endorsement effectiveness, J. Retailing Consum. Serv. 75 (2023) 103456, https://doi.org/10.1016/j.jretconser.2023.103456.
[62] E. Go, S.S. Sundar, Humanizing chatbots: the effects of visual, identity and conversational cues on humanness perceptions, Comput. Hum. Behav. 97 (2019) 304–316, https://doi.org/10.1016/j.chb.2019.01.020.
[63] K. Gray, D.M. Wegner, Feeling robots and human zombies: mind perception and the uncanny valley, Cognition 125 (2012) 125–130, https://doi.org/10.1016/j.cognition.2012.06.007.
[64] A. Prahl, L. Van Swol, Understanding algorithm aversion: when is advice from automation discounted? J. Forecast. 36 (2017) 691–702, https://doi.org/10.1002/for.2464.
[65] M. Bakpayev, T.H. Baek, P. van Esch, S. Yoon, Programmatic creative: AI can think but it cannot feel, Australas. Mark. J. 30 (2022) 90–95, https://doi.org/10.1016/j.ausmj.2020.04.002.
[66] J. Choi, S.R. Madhavaram, H.Y. Park, The role of hedonic and utilitarian motives on the effectiveness of partitioned pricing, J. Retailing 96 (2020) 251–265, https://doi.org/10.1016/j.jretai.2019.10.003.
[67] F. Faul, E. Erdfelder, A.G. Lang, A. Buchner, G*Power 3: a flexible statistical power analysis program for the social, behavioral, and biomedical sciences, in: Behav. Res. Methods, Psychonomic Society Inc., 2007, pp. 175–191, https://doi.org/10.3758/BF03193146.
[68] M. Perugini, M. Gallucci, G. Costantini, A practical primer to power analysis for simple experimental designs, Int. Rev. Soc. Psychol. 31 (2018), https://doi.org/10.5334/irsp.181.
[69] J. Cohen, Statistical power analysis for the behavioral sciences, Stat. Power Anal. Behav. Sci. 2nd (1988) 567, https://doi.org/10.1234/12345678.
[70] K.L. Nowak, C. Rauh, The influence of the avatar on online perceptions of anthropomorphism, androgyny, credibility, homophily, and attraction, J. Comput. Commun. 11 (2005) 153–178, https://doi.org/10.1111/j.1083-6101.2006.tb00308.x.
[71] V.L. Thomas, K. Fowler, Examining the impact of brand transgressions on consumers' perceptions of celebrity endorsers, J. Advert. 45 (2016) 377–390, https://doi.org/10.1080/00913367.2016.1172385.
[72] A. Alsaad, M.M. Alam, A. Lutfi, A sensemaking perspective on the association between social media engagement and pro-environment behavioural intention, Technol. Soc. 72 (2023) 102201, https://doi.org/10.1016/j.techsoc.2023.102201.
[73] J.S. Lim, J. Zhang, Adoption of AI-driven personalization in digital news platforms: an integrative model of technology acceptance and perceived contingency, Technol. Soc. 69 (2022) 101965, https://doi.org/10.1016/j.techsoc.2022.101965.
[74] Y.K. Choi, G.E. Miracle, F. Biocca, The effects of anthropomorphic agents on advertising effectiveness and the mediating role of presence, J. Interact. Advert. 2 (2001) 19–32, https://doi.org/10.1080/15252019.2001.10722055.
[75] T.Y. Wu, C.A. Lin, Predicting the effects of eWOM and online brand messaging: source trust, bandwagon effect and innovation adoption factors, Telematics Inf. 34 (2017) 470–480, https://doi.org/10.1016/J.TELE.2016.08.001.
[76] N.J. Evans, J. Phua, J. Lim, H. Jun, Disclosing Instagram influencer advertising: the effects of disclosure language on advertising recognition, attitudes, and behavioral intent, J. Interact. Advert. 17 (2017) 138–149, https://doi.org/10.1080/15252019.2017.1366885.
[77] E.A. Kim, D. Kim, E. Zihang, H. Shoenberger, The next hype in social media advertising: examining virtual influencers' brand endorsement effectiveness, Front. Psychol. 14 (2023), https://doi.org/10.3389/fpsyg.2023.1089051.
[78] B.B. McShane, E.T. Bradlow, J.G. Lynch, R.J. Meyer, EXPRESS: "Statistical significance" and statistical reporting: moving beyond binary, J. Mark (2023), https://doi.org/10.1177/00222429231216910.
[79] B. Liu, In AI we trust? Effects of agency locus and transparency on uncertainty reduction in Human–AI interaction, J. Comput. Commun. 26 (2021) 384–402, https://doi.org/10.1093/jcmc/zmab013.
[80] K.C. Yam, Y.E. Bigman, P.M. Tang, R. Ilies, D. De Cremer, H. Soh, K. Gray, Robots at work: people prefer—and forgive—service robots with perceived feelings, J. Appl. Psychol. 106 (2021) 1557–1572, https://doi.org/10.1037/apl0000834.
[81] B. Reeves, C. Nass, The Media Equation: How People Treat Computers, Television, and New Media like Real People and Places, Cambridge University Press, Cambridge, 1996. https://psycnet.apa.org/record/1996-98923-000.
[82] A. Purington, J.G. Taft, S. Sannon, N.N. Bazarova, S.H. Taylor, "Alexa is my new BFF": social roles, user satisfaction, and personification of the Amazon Echo, in: Conf. Hum. Factors Comput. Syst. - Proc., ACM, New York, NY, USA, 2017, pp. 2853–2859, https://doi.org/10.1145/3027063.3053246.
[83] J. Ham, S. Li, P. Shah, M.S. Eastin, The "Mixed" reality of virtual brand endorsers: understanding the effect of brand engagement and social cues on technological perceptions and advertising effectiveness, J. Interact. Advert. 23 (2023) 98–113, https://doi.org/10.1080/15252019.2023.2185557.
[84] I. Alnawas, J. Hemsley-Brown, The differential effect of cognitive and emotional elements of experience quality on the customer-service provider's relationship, Int. J. Retail Distrib. Manag. 46 (2018) 125–147, https://doi.org/10.1108/IJRDM-03-2017-0058.
[85] F. Deng, X. Jiang, Effects of human versus virtual human influencers on the appearance anxiety of social media users, J. Retailing Consum. Serv. 71 (2023) 103233, https://doi.org/10.1016/j.jretconser.2022.103233.
[86] M.M. Mariani, R. Perez-Vega, J. Wirtz, AI in marketing, consumer research and psychology: a systematic literature review and research agenda, Psychol. Market. 39 (2022) 755–776, https://doi.org/10.1002/mar.21619.
[87] F. Muniz, K. Stewart, L. Magalhães, Are they humans or are they robots? The effect of virtual influencer disclosure on brand trust, J. Consum. Behav. (2023), https://doi.org/10.1002/cb.2271.
[88] B. Melović, M. Jocović, M. Dabić, T.B. Vulić, B. Dudic, The impact of digital transformation and digital marketing on the brand promotion, positioning and

electronic business in Montenegro, Technol. Soc. 63 (2020) 101425, https://doi.org/10.1016/J.TECHSOC.2020.101425.
[89] A.D. Young, A.E. Monroe, Autonomous morals: inferences of mind predict acceptance of AI behavior in sacrificial moral dilemmas, J. Exp. Soc. Psychol. 85 (2019) 103870, https://doi.org/10.1016/j.jesp.2019.103870.
[90] S. Sands, C. Ferraro, V. Demsar, G. Chandler, False idols: unpacking the opportunities and challenges of falsity in the context of virtual influencers, Bus. Horiz. 65 (2022) 777–788, https://doi.org/10.1016/j.bushor.2022.08.002.
[91] M.-H. Huang, R.T. Rust, A strategic framework for artificial intelligence in marketing, J. Acad. Market. Sci. 49 (2021) 30–50, https://doi.org/10.1007/s11747-020-00749-9.
[92] G. Löhr, Do socially disruptive technologies really change our concepts or just our conceptions? Technol. Soc. 72 (2023) 102160 https://doi.org/10.1016/j.techsoc.2022.102160.
[93] J.H. Cheung, D.K. Burns, R.R. Sinclair, M. Sliter, Amazon Mechanical Turk in organizational psychology: an evaluation and practical recommendations, J. Bus. Psychol. 32 (2017) 347–361, https://doi.org/10.1007/s10869-016-9458-5.
[94] R. De Cicco, S.C.L. da Costa e Silva, R. Palumbo, Should a chatbot disclose itself? Implications for an online conversational retailer, in: Lect. Notes Comput. Sci. (Including Subser. Lect. Notes Artif. Intell. Lect. Notes Bioinformatics), Springer Science and Business Media Deutschland GmbH, 2021, pp. 3–15, https://doi.org/10.1007/978-3-030-68288-0_1.
[95] M. Shin, S.J. Kim, F. Biocca, The uncanny valley: No need for any further judgments when an avatar looks eerie, Comput. Hum. Behav. 94 (2019) 100–109, https://doi.org/10.1016/j.chb.2019.01.016.

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